CORE LIFTING®

WIRE ROPE

BRIDON

steel wire ropes by BEKAERT

Dyform 34LR
Dyform 28HML
Dyform 18
Dyform 6
Dyform 8
Constructex
6x19 Class
6x37 Class

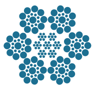
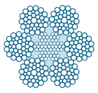
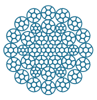
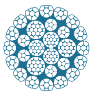
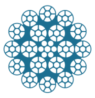
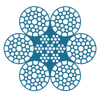
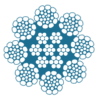
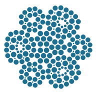
Bekaert

Bridon®

Crane & Industrial

Quick Reference Guide

Wire Rope Application Guide for Cranes

mm	Standard		High Performance					
	6x19 	6x36 	Dyform 34LR 	Dyform 28 HML 	Dyform 18 	Dyform 6 	Dyform 8 	Constructex 
Telescopic Mobile Crane								
Auxiliary Rope	●	●	●	●	●			
Main Hoist Line			●	●	●			
Tower Crane								
Main Hoist Rope			●	●				
Derricking Rope	●	●				●	●	
Trolley Rope	●	●				●	●	
Container Crane								
Main Hoist Rope	●	●				●	●	
Boom Hoist Rope	●					●	●	
Trolley Rope	●	●				●	●	
Mobile Lattice Boom Crane								
Main Hoist Rope			●	●	●			
Boom Hoist Rope	●					●	●	●
Auxiliary Rope			●	●	●			
Dockside Crane								
Main Hoist Rope			●	●	●			
Boom Hoist Rope	●					●		●
Overhead Crane								
Main Hoist Rope	●	●				●	●	
Steel Mill Ladle Crane								
Main Hoist Rope	●	●				●	●	
Offshore Pedestal Crane								
Main Hoist Rope			●	●	●			
Whipline Rope			●	●	●			
Boom Hoist Rope	●					●	●	●
Unloader Crane								
Main Hoist Rope	●	●				●		●
Boom Hoist Rope	●	●				●		●
Closing Rope	●	●				●		●
Racking Rope		●				●		
Piling Crane								
Main Hoist Rope					●			

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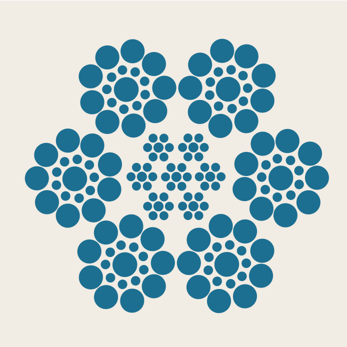
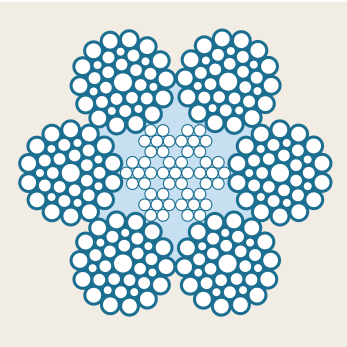
Standard Hoist Wire Ropes

6 X 19
Classification



Diameter	Approx Mass		Minimum Breaking Force			
	IWRC		EIP IWRC		EEIP IWRC	
in	lb/ft	kg/m	tons	kN	tons	kN
1/4	0.108	0.16	3.40	30.2		
5/16	0.169	1.74	5.27	46.9		
3/8	0.244	0.36	7.55	67.2		
7/16	0.332	0.49	10.2	90.7	11.2	100
1/2	0.434	0.65	13.3	118	14.6	130
9/16	0.549	0.82	16.8	149	18.5	165
5/8	0.688	1.02	20.6	183	22.7	202
3/4	0.975	1.45	29.4	262	32.4	288
7/8	1.33	1.98	39.8	354	43.8	390
1	1.73	2.57	51.7	460	56.9	506
1 1/8	2.19	3.26	65.0	578	71.5	636
1 1/4	2.75	4.09	79.9	711	87.9	782
1 3/8	3.28	4.88	96.0	854	106	943
1 1/2	3.90	5.80	114	1,01	125	1,11

6 X 37
Classification



Diameter	Approx Mass		Minimum Breaking Force			
	IWRC		EIP IWRC		EEIP IWRC	
in	lb/ft	kg/m	tons	kN	tons	kN
1/4*	0.108	0.16	3.40	30.3		
5/16*	0.169	1.74	5.27	46.9		
3/8*	0.244	0.36	7.55	67.2		
7/16*	0.332	0.49	10.2	90.7	11.2	100
1/2	0.434	0.65	13.3	118	14.6	130
9/16	0.549	0.82	16.8	149	18.5	165
5/8	0.688	1.02	20.6	183	22.7	202
3/4	0.975	1.45	29.4	262	32.4	288
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1 1/4	2.75	4.09	79.9	711	87.9	782
1 3/8	3.28	4.88	96.0	854	106	943
1 1/2	3.90	5.80	114	1,01	125	1,110

Note: all sizes Powerchecked and mass calculation is approximate.
* Asterisk indicates nonstandard item

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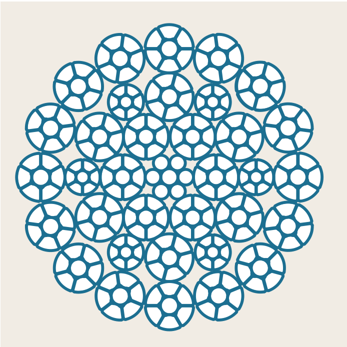
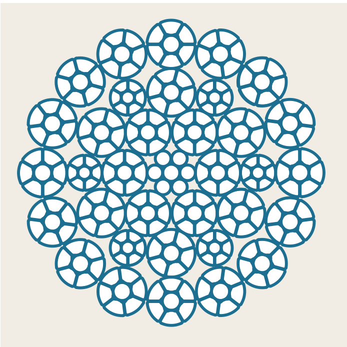
High Performance Main/Auxiliary Hoist Wire Ropes

Dyform® 34LR/PI



Diameter		Approx Mass		Minimum Breaking Force Rope Grade			
		WSC		EIP/1960		EEIP/2160	
in	mm	lb/ft	kg/m	kN	tons	kN	tons
	14	0.659	0.980	179	20.1	191	21.5
9/16		0.687	1.02	185	20.8	201	22.6
5/8		0.860	1.28	232	26.1	251	28.2
	16	0.860	1.28	232	26.1	251	28.2
	18	1.09	1.62	298	33.5	319	35.9
	19	1.22	1.81	331	37.2	356	40.0
3/4		1.22	1.81	331	37.2	356	40.0
	20	1.34	2.00	370	41.6	397	44.6
	22	1.63	2.42	442	49.7	482	54.2
7/8		1.63	2.42	448	50.4	487	54.7
	24	1.94	2.88	528	59.3	569	64.0
1		2.17	3.23	586	65.9	623	70.0
	26	2.27	3.38	618	69.5	660	74.2
	28	2.63	3.92	712	80.0	758	85.2
1 1/8		2.75	4.09	743	83.5	779	87.6
	30	3.02	4.50	823	92.5	857	96.3
1 1/4		3.44	5.12	919	103	1010	113

Dyform® 34XL



Diameter		Approx Mass		Minimum Breaking Force Rope Grade	
		WSC		XL	
in	mm	lb/ft	kg/m	tons	kN
5/8	16	0.84	1.25	30.6	272
3/4	19	1.31	1.95	42.9	382
1		2.15	3.20	71.7	638
	26	2.28	3.39	74.0	658
	28	2.63	3.91	84.4	751
1 1/8		2.72	4.05	86.9	773
	29	2.94	4.38	86.9	829

Note: all sizes Powerchecked and mass calculation is approximate.
* Asterisk indicates nonstandard item

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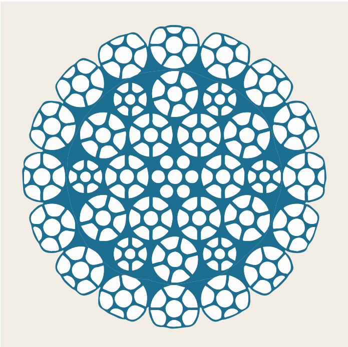
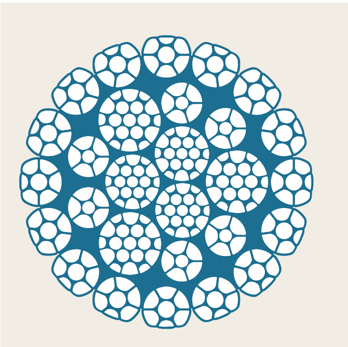
Dyform® 34MAX



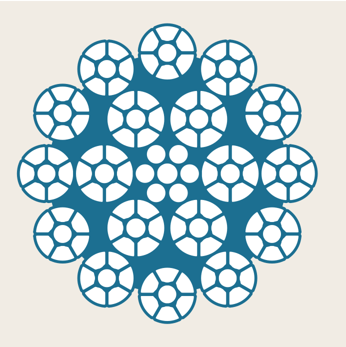
Diameter		Approx Mass WSC		Minimum Breaking Force Rope Grade MAX	
in	mm	lb/ft	kg/m	tons	kN
1		2.22	3.30	76.9	684
	26	2.38	3.54	79.3	705
	28	2.86	4.26	91.9	818
1 1/8		3.06	4.55	95.3	848
1 1/4	32	3.74	5.57	122	1,085

Dyform® 28HML



Diameter		Approx Mass WSC		Minimum Breaking Force Rope Grade MAX	
in	mm	lb/ft	kg/m	tons	kN
	15	0.814	1.21	26.9	239
	17	1.06	1.58	34.5	307
	18	1.20	1.79	38.7	344
	20	1.57	2.34	47.7	424
	21	1.64	2.44	52.6	468
7/8	22	1.88	2.80	58.9	524
	23	2.06	3.07	63.1	561
	25	2.41	3.59	74.5	663
	26	2.59	3.85	79.2	705
	28	2.85	4.24	91.5	814
1 1/8		2.90	4.32	95.3	848
	30	3.06	4.55	105	935
1 1/4	32	3.37	5.02	122	1,085

Dyform®
18/18PI

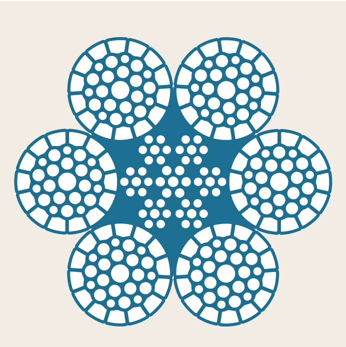


Diameter		Approx Mass WSC		Minimum Breaking Force Rope Grade EEIP/2160	
in	mm	lb/ft	kg/m	tons	kN
3/8		0.281	0.42	8.31	73.9
	10	0.336	0.50	9.49	84.4
	11	0.407	0.61	11.7	104
7/16		0.415	0.62	11.2	100
	12	0.484	0.72	13.7	122
1/2		0.542	0.81	14.6	130
	13	0.568	0.85	16.1	143
	14	0.659	0.98	18.5	165
9/16		0.686	1.02	19.3	172
5/8	16	0.847	1.26	22.7	202
	18	1.09	1.62	30.8	274
3/4	19	1.22	1.82	32.4	288
	20	1.34	1.99	37.9	337
	22	1.63	2.43	46.8	416
7/8		1.66	2.47	46.8	416
	24	1.94	2.89	54.6	486
1		2.17	3.23	57.5	512
	26	2.27	3.38	64.1	570
	28	2.63	3.91	74.3	661
1 1/8		2.74	4.08	71.5	636
1 1/4	32	3.39	5.04	87.9	782

Note: all sizes Powerchecked and mass calculation is approximate.
* Asterisk indicates nonstandard item

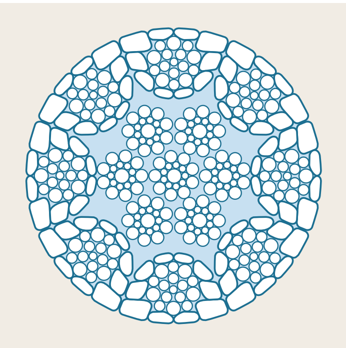
High Performance Boom Hoist Wire

Dyform®
6/6PI



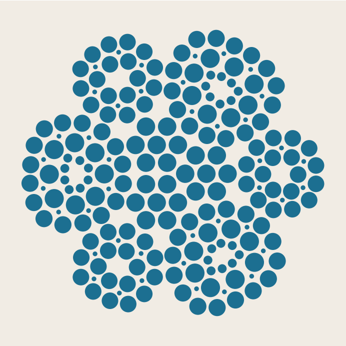
Diameter		Approx Mass WSC		Minimum Breaking Force Rope Grade EEIP/2160	
in	mm	lb/ft	kg/m	tons	kN
3/8		0.285	0.42	8.79	78.2
	10	0.308	0.46	9.69	86.2
	11	0.373	0.56	11.9	106
7/16		0.376	0.56	11.9	106
	12	0.444	0.66	13.9	124
1/2		0.497	0.74	15.3	136
	13	0.521	0.78	16.0	142
	14	0.605	0.90	18.5	165
9/16		0.633	0.94	19.3	172
5/8	16	0.775	1.15	23.6	210
	18	1.00	1.49	30.1	268
3/4	19	1.10	1.49	32.4	288
	20	1.23	1.83	37.2	331
	22	1.47	2.19	45.01	401
7/8		1.52	2.26	45.01	401
	24	1.78	2.65	53.6	477
1		1.92	2.86	57.5	512
	26	2.07	3.08	62.9	560
	28	2.36	3.51	73.0	649
1 1/8		2.54	3.78	76.0	676
1 1/4	32	3.13	4.66	87.9	782
1 3/8		3.79	5.64	106	943
1 1/2	38	4.00	5.95	125	1113

Dyform®
8 MAX PI



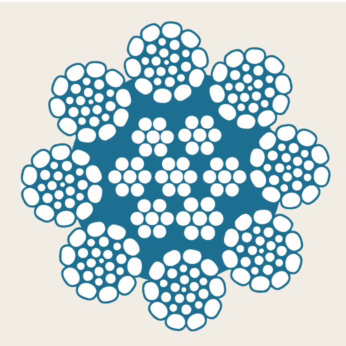
Diameter		Approx Mass WSC		Minimum Breaking Force MAX	
in	mm	lb/ft	kg/m	tons	kN
	24.0	1.96	2.92	61.1	544
1		2.15	3.20	64	569
	26.0	2.27	3.38	71.8	639
	28.0	2.64	3.93	83.3	741
1 1/8*		2.75	4.09	86.9	773
	30.0	3.03	4.51	95.7	851
1 1/4	32	3.65	5.43	109	970

Constructex®



Diameter	Approx Mass WSC		Minimum Breaking Force	
in	lb/ft	kg/m	tons	kN
5/8*	0.900	1.34	25.5	227
3/4*	1.1	1.64	36.5	325
7/8*	1.5	2.23	48.5	432
1*	2.0	2.98	62.5	556
1 1/8*	2.6	3.87	79.5	707
1 1/4*	3.2	4.76	97.6	868
1 3/8	3.9	5.80	119	1059
1 1/2	4.7	6.99	139	1237
1 5/8	5.7	8.48	162	1441
1 3/4*	6.2	9.23	185	1,650

Dyform®
8/8PI



Diameter		Approx Mass WSC		Minimum Breaking Force EIP/1960	
in	mm	lb/ft	kg/m	tons	kN
3/8		0.306	0.46	9.69	86.2
	10*	0.316	0.47	10.0	89.2
	11*	0.383	0.57	12.4	110
7/16		0.391	0.58	12.4	110
	12*	0.456	0.68	14.4	128
1/2		0.505	0.75	16.2	144
	13*	0.535	0.80	16.9	150
	14*	0.620	0.92	19.6	174
9/16*		0.646	0.96	20.3	181
5/8	16	0.825	1.23	25.0	222
	18*	1.03	1.53	32.1	286
3/4	19	1.19	1.77	35.7	318
	20	1.25	1.86	36.7	353
	22	1.48	2.20	48.0	427
7/8		1.53	2.28	48.0	427
	24	1.76	2.62	58.2	517
1		2.04	3.04	62.8	559
	26*	2.14	3.18	67.0	596
	28	2.37	3.53	77.7	691
1 1/8		2.68	3.99	81.8	727
1 1/4	32	3.16	4.70	102	907
1 3/8		4.11	6.12	129	1148
1 1/2	38	4.64	6.91	138	1,228

Note: all sizes Powerchecked and mass calculation is approximate.
* Asterisk indicates nonstandard item

Constructex For Oilfield Applications

Constructex is designed with nine strands of three different strand constructions. Each outside strand is manufactured with a polymer center to give the rope added flexibility in handling. The wire rope is swaged as a final operation to postform the rope and to provide a smooth outer surface due to the triangular shape of the strands. Constructex is available in diameters from 5/8" through 1 5/8".

Applications

Constructex is recommended as a drill line, tubing line and boom hoist rope where longer service life is desired. Superior abrasion and crushing resistance makes Constructex the ideal rope for these applications. Constructex can be recommended with confidence wherever abusive conditions exist, particularly where the rope is retired because of broken wires resulting from wear and crushing on the drum.

Characteristics

Higher Strength:

The compacted design of Constructex allows the rope to have a greater metallic area and a higher strength than conventional ropes. This provides the end user the opportunity to lift heavier loads with the same or even smaller rope diameter. For example, a 1" Constructex has a minimum breaking force of 62.5 tons, compared to 56.9 tons for a 1" 6x19 EEIPS and 51.7 tons for 6x19 EIPS.

Increased Crush Resistance:

The unique design of Constructex, with single operation closing and compaction, results in a smooth exterior surface, a reduction of internal voids and excellence in resisting damage from crushing.

Longer Rope Life:

Constructex ropes will provide longer rope life in many abusive applications due to the combination of higher strength and improved resistance to crushing. As with all ropes, proper daily inspection of the rope and equipment is necessary to assure the user that a safe operation is maintained. For the user, longer rope life means cost savings due to fewer rope changes and less down time.

Inspection of Wire Rope

Introduction

The most important aspect of operating a rope safely in oilfield applications is properly performed inspection, done on a regular basis. The American Petroleum Institute (API) publishes a series of Specifications and Recommended Practices that provide information on wire rope, use of wire rope, detailed inspection procedures and retirement criteria. The applicable standard for wire rope is API 9A and recommended practices are RP 2D, RP 4G, RP 9B and RP 54. A summation of these standards and practices is provided in this document. For more detailed information refer to the specific standard or recommendation practice.

Inspection

All hoisting lines should be visually inspected once each working day and ideally, should cover all rope expected to be in use during operations on that day. The inspection must be more than just a quick look. It needs to be done carefully and in enough light to find damage or broken wires that may require the rope to be taken out of service. It must also be remembered that a dirty or greasy rope is almost impossible to inspect properly, as dirt and grease may hide problem areas.
The individual making the inspection should be familiar with the machine, the wire rope, and that particular application.

Special care should always be taken when inspecting common repetitive wear sections such as:

(1) Flange step up, cross over and repetitive pick up points on the drum
(2) Areas of the rope operating through a reverse bend in the reeving system
(3) Equalizer sheaves
(4) End connections

The inspector should be concerned with discovering gross damage that may be an immediate hazard. Specific types of damage include the following:

(1) Distortion to the uniform structure of the rope
(2) Broken wires
(3) Corrosion
(4) Gross damage to, or deterioration of, end connections
(5) Evidence of heat/electrical/lightning damage
(6) Localized change in lubrication condition

When damage is discovered, the affected sections must be evaluated as detailed in the replacement section below to determine if the rope needs to be removed from service.

Details of daily inspections do not need to be documented, but it is a good idea to keep a frequent inspection log on the crane, simply noting time, date and identity of the inspector.

Periodically, a detailed inspection needs to be performed and documented. The inspection frequency needs to be based on factors such as expected rope life as determined by experience on the particular installation or similar installations, severity of environment, percentage of capacity lifts, frequency rates of operation, and exposure to shock loads. Inspections need not be at equal calendar intervals and should be more frequent as the rope approaches the end of its useful life. There are many duty cycle rope applications where the service life is less than a month, or sometimes even a week in severe service conditions, so a periodic level of inspection may have to be performed daily.

The periodic inspection must cover the surface of the entire rope length and no attempt should be made to open the rope. In addition to common repetitive wear sections checked during the frequent inspection, additional sections prone to rapid deterioration such as the following need special attention.

(1) Locations where rope vibrations are damped, such as the following:
    a. Sections in contact with equalizer sheaves, or other sheaves where rope travel is limited
    b. Sections of the rope at or near end connections where corroded or broken wires may protrude
(2) Bridle reeving in the boom hoist ropes
(3) Repetitive pickup points and crossover and change of layer points at flanges on drums
(4) Fleeting or deflector sheaves

In addition to the specific types of damage listed in the frequent inspection section, these additional items need to be addressed:

(1) Measuring the rope diameter in numerous locations to assess uniform loss of diameter along the entire length of rope
(2) Close visual observation of the entire length to identify
    a. Lengthening of lay in localized areas
    b. Diameter reduction in localized areas
    c. Distortion of rope structure (kinking, birdcaging, crushing)
    d. Steel core protrusion between the outer strands
    e. Internal corrosion
    f. Wear of outside wires
    g. More detailed inspection of end connections for broken wires and corrosion
    h. Severely corroded, cracked, bent, worn or improperly applied end connections
    i. Waviness (corkscrew effect) of rope
    j. High or low strand

To establish data as a basis for judging the proper time for replacement, a dated report of rope condition at each periodic inspection must be kept on file. This report shall cover points of deterioration listed above. If the rope is replaced, only the fact that the rope was replaced need be recorded.

Certain types of ropes and applications require special attention and require reduced time intervals between periodic inspections:

- Rotation Resistant ropes have a unique construction and are susceptible to damage and increased deterioration when working under difficult conditions such as duty cycle operation.
- Boom hoist ropes because of the importance of their function and because their location may make inspection difficult.

Rope Replacement

There are no precise rules to determine the exact time for the replacement of the rope since many variable factors are involved. Once a rope reaches any one of the removal criteria, it must be replaced immediately unless allowed to operate to the end of the work shift by the judgment of a qualified person. If the rope was not removed immediately, it shall be replaced before the equipment is returned to service.

Conditions such as the following are sufficient reason to require the rope to be removed from service:

(1) Broken wires:
    a. In running ropes (6 and 8 strand constructions, expect 6x7 and rotation resistant ropes) 6 randomly distributed wire breaks per rope lay or 3 wire breaks per strand per rope lay. A rope lay is the distance that it takes one outer strand to make one complete revolution around the rope. A 6 strand rope will typically have a rope lay of 6.4 times the rope diameter (i.e. A 1/2" 6x25FW EIP IWRC RRL rope will have a rope lay of 3.2").
    b. For all categories of Rotation Resistant ropes the retirement criteria is 2 wire breaks in 6 rope diameters or 4 wire breaks in 30 rope diameters (i.e. 6 rope diameters in a 1" rope is 6")
    c. In 6x7 running ropes 3 broken wires in one lay length
    d. Two broken outer wires at strand to strand or strand to core contact point (valley breaks)
    e. In standing ropes, more than 3 broken wires in one lay away from the end connection or more than 2 broken wires at the end connection
(2) Reductions from nominal diameter greater than 5% (Minimum Value = Nominal Diameter x .95)
(3) Distortion of rope structure:
    a. Damage resulting in distortion of the rope structure (e.g., kinking, birdcaging, crushing)
    b. Steel core protrusion between the outer strands
    c. Localized change in lay length
    d. Changes in original geometry due to crushing forces where the diameter across the distorted section is 5/6 of the nominal diameter
(4) Waviness (corkscrew effect) in the rope that causes overall diameter to increase to a value greater than 110% of nominal rope diameter
(5) A high or low strand that is higher or lower than 1/2 of the strand diameter above or below the surface of the rope
(6) Any apparent damage from a heat source including, but not limited to welding, power line strikes, or lightning
(7) Widespread or localized external corrosion as evidenced by pitting, and obvious signs of internal corrosion such as magnetic debris coming from valleys
(8) Severely corroded, cracked, bent, worn, grossly damaged, or improperly installed end connections

Rope Service Life

A long-range inspection program should be established and should include records on the examination of ropes removed from service so that a relationship can be established between visual observation and actual condition of the internal structure.

There are a wide variety of wire rope constructions available to be used on cranes. It is important that the correct rope be used for each specific application. Because wire rope wears in service, the method by which the rope wears is an important factor in determining the most suitable rope. Replacement rope must have a rated strength at least equal to the original rope supplied or recommended for the machine. Any change from the original specification for the rope must be specified by the wire rope manufacturer, crane manufacturer, or qualified person. When there is a question, consult with Bridon American about the rope construction most appropriate for the application.

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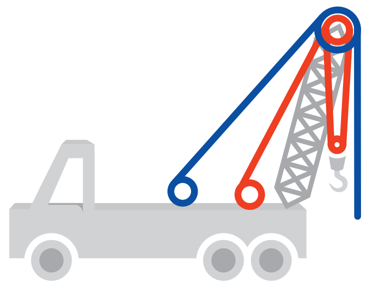
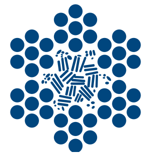
Integrity in Every Strand

Quick Reference Guide: Oil Field Ropes

Well Servicing Rig

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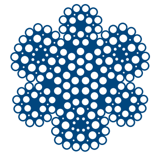
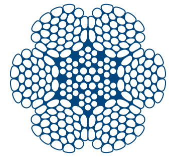
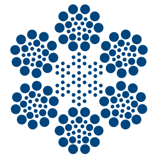
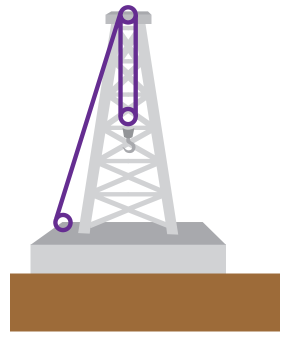
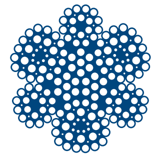

Constructex


Swaged Tubing Line


Tubing Line

Rotary Drilling Rig

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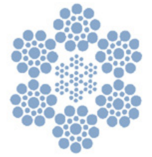
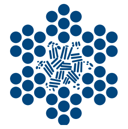

Drill Line

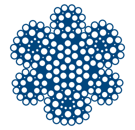


Sand Line

- Large outer wires for resistance to wear.
- Exceptional spooling characteristics.
- Resistance to kinking.
- Easy to splice.

Diameter	Approx. Mass WSC		Min. Breaking Force	
in	lb/ft	kg/ft	tons	kN
3/8	0.21	0.09	5.90	52.20
7/16	0.29	0.13	7.90	70.60
1/2	0.38	0.17	10.30	91.70
9/16	0.48	0.21	13.00	115.70
5/8	0.59	0.26	15.90	141.50
3/4	0.84	0.37	22.70	202.00

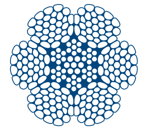
NOTE: Typical Construction 6x7 Fiber Core.



Constructex

- Swaged to increase wearing surface and density.
- Long service life due to resistance to scrubbing and crushing.
- High breaking force.
- Flexible construction.

Diameter	Approx. Mass WSC		Min. Breaking Force	
			EIP	
in	lb/ft	kg/ft	tons	kN
5/8	0.90	0.39	25.50	226.90
3/4	1.10	0.50	36.50	324.70
7/8	1.50	0.68	48.50	431.50
1	2.00	0.91	62.50	556.00
1 1/8	2.60	1.18	79.50	707.30
1 1/4	3.20	1.45	97.60	868.30
1 3/8	3.80	1.72	119.00	1058.70
1 1/2	4.60	2.09	139.00	1236.70
1 5/8	5.30	2.41	162.00	1441.30

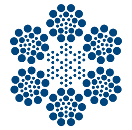


Swaged Tubing Line

- Excellent resistance to crushing.
- High breaking force.
- Good resistance to abrasion.

Diameter	Approx. Mass WSC		Min. Breaking Force	
in	lb/ft	kg/ft	tons	kN
7/8	1.70	0.77	47.40	421.90
1	2.22	1.01	62.00	551.80
1 1/8	2.80	1.27	73.50	654.20

NOTE: Typical Construction 6x31 Swaged.

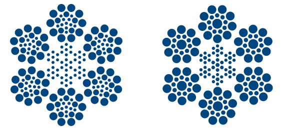


Tubing Line

- Combination of flexibility and resistance to crushing.
- Outstanding resistance to wear and fatigue.

Diameter	Approx. Mass WSC		Min. Breaking Force	
			EIP	
in	lb/ft	kg/ft	tons	kN
3/4	1.04	0.46	29.40	261.70
7/8	1.41	0.62	39.80	354.20
1	1.85	0.82	51.70	460.10
1 1/8	2.34	1.03	65.00	578.50

NOTE: Typical Construction 6x26 IWRC.

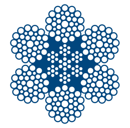


Rotary Drill Line

- Combination of strength, flexibility and resistance to peening.
- Good resistance to wear and fatigue.
- Long service life when sheaves and drums are of moderate size.

Diameter	Approx. Mass WSC		Min. Breaking Force			
			EIP		EEIP	
in	lb/ft	kg/ft	tons	kN	tons	kN
1	1.85	0.82	51.70	460.10	56.90	506.40
1 1/8	2.34	1.03	65.00	578.50	71.50	636.40
1 1/4	2.89	1.28	79.90	711.10	87.90	782.30
1 3/8	3.49	1.54	96.00	854.40	106.00	943.40
1 1/2	4.16	1.84	114.00	1014.60	125.00	1112.50
1 5/8	4.88	2.15	132.00	1174.80	146.02	1299.40
1 3/4	5.66	2.50	153.00	1361.70	169.00	1504.10

Note: Typical Construction 6x19(S) & 6x26(WS).

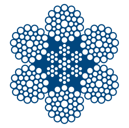


6x19 Class IWRC

- High quality 6-strand rope is exceptionally robust.
- Superior strength confirmed by Bridon's Powercheck process for testing a sample.
- Durability and outstanding resistance to wear.
- Fully lubricated to reduce wear and tear.

Diameter	Approx. Mass WSC		Min. Breaking Force			
			EIP IWRC		EEIP IWRC	
in	lb/ft	kg/ft	tons	kN	tons	kN
1/4	0.12	0.05	3.40	30.30	—	—
5/16	0.18	0.08	5.27	46.90	—	—
3/8	0.26	0.11	7.55	67.20	—	—
7/16	0.35	0.15	10.20	90.70	11.20	99.60
1/2	0.46	0.20	13.30	118.40	14.60	129.90
9/16	0.58	0.26	16.80	149.50	18.50	164.70
5/8	0.72	0.32	20.60	183.30	22.70	202.00
3/4	1.04	0.46	29.40	261.70	32.40	288.40
7/8	1.41	0.62	39.80	354.20	43.80	389.80
1	1.85	0.82	51.70	460.10	56.90	506.40
1 1/8	2.34	1.03	65.00	578.50	71.50	636.40
1 1/4	2.89	1.28	79.90	711.10	87.90	782.30
1 3/8	3.49	1.54	96.00	854.40	106.00	943.40
1 1/2	4.16	1.84	114.00	1014.60	125.00	1112.50

NOTE: Metric rope diameters on request. Check www.bridonamerican.com or call customer service for more information.
NOTE: Sizes 1/4" to 1" Powerchecked.
NOTE: Tuggers.

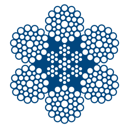


6x36 Class IWRC

- High quality 6-strand rope with increased flexibility.
- Superior strength confirmed by Bridon's Powercheck process for testing a sample (where noted in the table).
- Flexible construction with good bend-fatigue life in most applications.
- Fully lubricated to reduce wear and tear.

Diameter	Approx. Mass WSC		Min. Breaking Force			
			EIP IWRC		EEIP IWRC	
in	lb/ft	kg/ft	tons	kN	tons	kN
1/4	0.12	0.05	3.40	30.30	—	—
5/16	0.18	0.08	5.27	46.90	—	—
3/8	0.26	0.12	7.55	67.20	—	—
7/16	0.35	0.16	10.20	90.70	11.20	99.60
1/2	0.46	0.20	13.30	118.40	14.60	129.90
9/16	0.58	0.26	16.80	149.50	18.50	164.70
5/8	0.72	0.32	20.60	183.30	22.70	202.00
3/4	1.04	0.46	29.40	261.70	32.40	288.40
7/8	1.41	0.62	39.80	354.20	43.80	389.80
1	1.85	0.82	51.70	460.10	56.90	506.40
1 1/8	2.34	1.03	65.00	578.50	71.50	636.40
1 1/4	2.89	1.28	79.90	711.10	87.90	782.30
1 3/8	3.49	1.54	96.00	854.40	106.00	943.40
1 1/2	4.16	1.84	114.00	1014.60	125.00	1112.50
1 5/8	4.88	2.15	132.00	1174.80	146.00	1299.40
1 3/4	5.66	2.50	153.00	1361.70	169.00	1504.10
1 7/8	6.49	2.86	174.00	1548.60	192.00	1708.80
2	7.39	3.26	198.00	1762.20	217.00	1931.30
2 1/8	8.34	3.68	221.00	1966.90	243.00	2162.70
2 1/4	9.35	4.13	247.00	2198.30	272.00	2420.80
2 3/8	10.42	4.60	274.00	2438.60	301.00	2678.90

NOTE: Metric rope diameters on request. Check www.bridonamerican.com or call customer service for more information.
NOTE: Sizes 1/4" to 1" Powerchecked.
NOTE: Winch lines, mast raising lines.



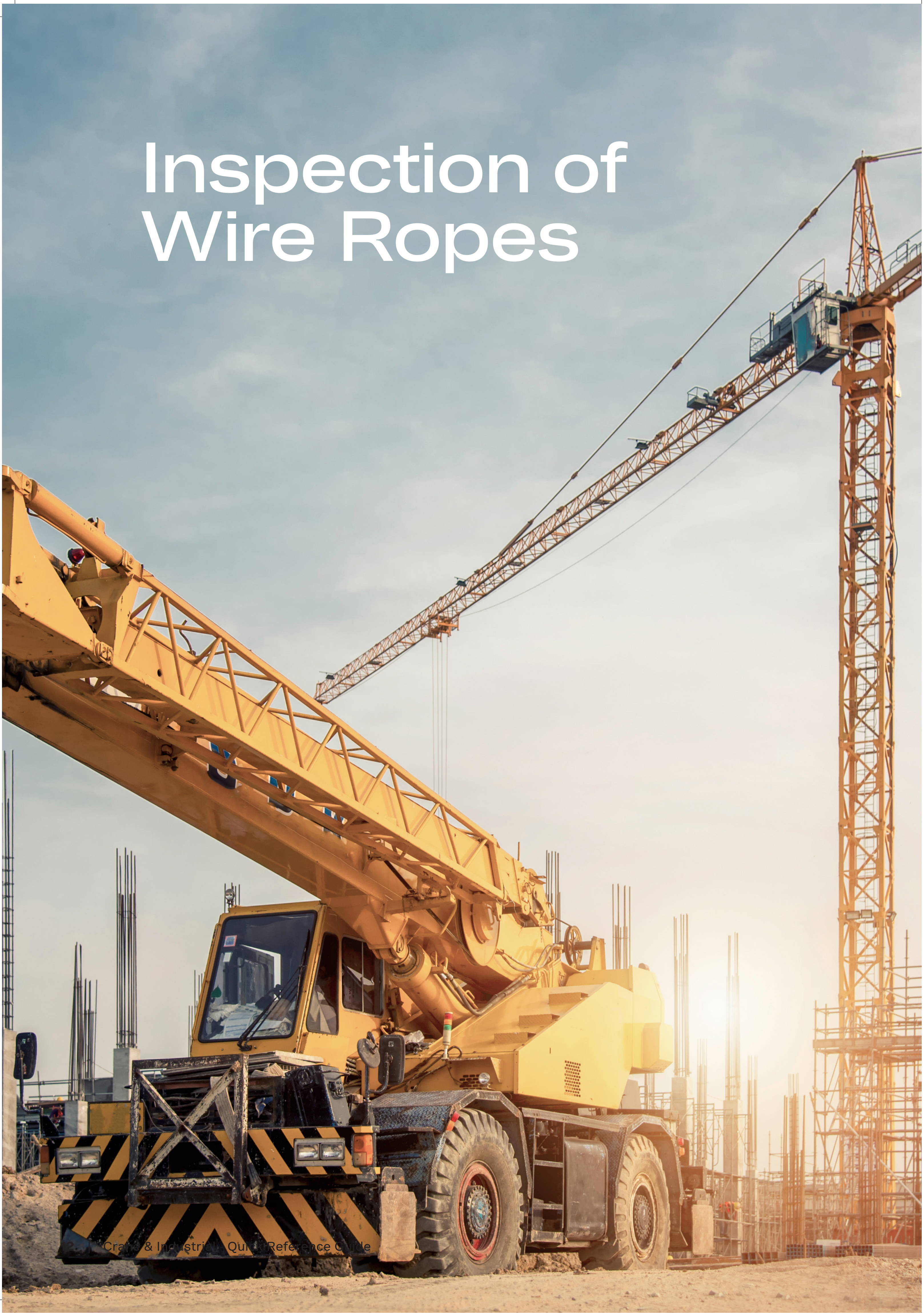
6x19 / 6x36 / 6x61 Classes IWRC

- High quality 6-strand rope with increased flexibility.
- Superior strength confirmed by Bridon's Powercheck process for testing a sample (where noted in the table).
- Flexible construction with good bend-fatigue life in most applications.
- Fully lubricated to reduce wear and tear.

Diameter	Approx. Mass		Min. Breaking Force					
			6x19 IWRC		6x36 IWRC		6x61 IWRC	
in	lb/ft	kg/ft	tons	kN	tons	kN	tons	kN
2 1/2	11.60	5.12	302.00	2687.80	302.00	2687.80	—	—
2 5/8	12.80	5.64	331.00	2945.90	331.00	2945.90	—	—
2 3/4	14.00	6.19	361.00	3212.90	361.00	3212.90	—	—
2 7/8	15.30	6.77	—	—	392.00	3488.80	—	—
3	16.60	7.32	—	—	425.00	3782.50	—	—
3 1/8	18.00	7.96	—	—	458.00	4076.20	—	—
3 1/4	19.50	8.60	—	—	492.00	4378.80	—	—
3 3/8	21.00	9.27	—	—	529.00	4708.10	—	—
3 1/2	22.70	10.03	—	—	564.00	5019.60	555.00	4939.50
3 5/8	24.30	10.73	—	—	602.00	5357.80	592.00	5268.80
3 3/4	26.00	11.49	—	—	641.00	5704.90	632.00	5624.80
3 7/8	27.70	12.22	—	—	680.00	6052.00	669.00	5954.10
4	29.60	13.08	—	—	720.00	6408.00	713.00	6345.70
4 1/8	31.40	13.87	—	—	761.00	6772.90	753.00	6701.70
4 1/4	33.40	14.75	—	—	803.00	7146.70	799.00	7111.10
4 3/8	35.40	15.64	—	—	846.00	7529.40	846.00	7529.40
4 1/2	37.40	16.52	—	—	889.00	7912.10	889.00	7912.10
4 5/8	39.50	17.43	—	—	934.00	8312.60	934.00	8312.60
4 3/4	41.70	18.41	—	—	979.00	8713.10	979.00	8713.10

NOTE: Metric rope diameters on request. Check www.bridonamerican.com or call customer service for more information.
NOTE: Sizes 1/4" to 1" Powerchecked.
NOTE: Winch lines, mast raising lines.

Inspection of Wire Ropes



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INTRODUCTION

The most important aspect of operating a rope safely is regular proper inspection. ASME crane safety standards such as B30.2 and B30.5 provide detailed inspection procedures and retirement criteria. Both standards specify that all running ropes in service should be visually inspected once each working day and shall consist of observation of all rope that can reasonably be expected to be in use during operations on that day. The inspection must be more than just a quick look. It needs to be done carefully and in enough light to find damage or broken wires that may require the rope to be taken out of service. It must also be remembered that a dirty or greasy rope is almost impossible to inspect properly, as dirt and grease may hide problem areas. The individual making the inspection should be familiar with the machine, the wire rope, and that particular application. The B30 standards provide information on both a frequent inspection to be done daily and a much more detailed periodic inspection that is done on a weekly or monthly basis.

FREQUENT INSPECTION

As stated previously, all running ropes in service should be visually inspected once each working day and shall consist of observation of all rope that can reasonably be expected to be in use during operations on that day. The inspector should know where and how rope on the particular application wears out so that the daily inspection can be focused on the known wear areas. Special care should always be taken when inspecting common repetitive wear sections such as:

Flange step up, cross over points and repetitive pick up on the drum; areas of the rope operating through a reverse bend in the reeving system, equalizer sheaves, and end connections.

The inspector should be concerned with discovering gross damage that may be an immediate hazard. Specific types of damage include the following:

Distortion to the uniform structure of the rope; broken wires; corrosion, gross damage to or deterioration of end connections, evidence of heat/electrical/lightning damage, and localized change in lubrication condition.

When damage is discovered, a qualified person must evaluate affected sections as detailed in the rope replacement section below to determine if the rope needs to be removed from service. The B30 standards do not require frequent inspections to be documented, but it is a good idea to keep a frequent inspection log on the crane, simply noting time, date and identity of the inspector.

PERIODIC INSPECTION

The inspection frequency needs to be based on factors such as expected rope life as determined by experience on the particular installation or similar installations, severity of environment, percentage of capacity lifts, frequency rates of operation, and exposure to shock loads. Inspections need not be at equal calendar intervals and should be more frequent as the rope approaches the end of its useful life. There are many duty cycle rope applications where the service life is less than a month, or sometimes even a week in severe service conditions, so a periodic level of inspection may have to be performed daily.

The periodic inspection must cover the surface of the entire rope length and no attempt should be made to open the rope. In addition to common repetitive wear sections checked during the frequent inspection, additional sections prone to rapid deterioration such as the following need special attention.

(1) Locations where rope vibrations are damped, such as the following: sections in contact with equalizer sheaves, or other sheaves where rope travel is limited; sections of the rope at or near end connections where corroded or broken wires may protrude; bridle reeving in the boom hoist ropes; repetitive pickup points and crossover and change of layer points at flanges on drums; fleeting or deflector sheaves.

In addition to the specific types of damage listed in the frequent inspection section, these additional items need to be addressed: Measuring the rope diameter in numerous locations to assess uniform loss of diameter along the entire length of rope; close visual observation of the entire length to identify; lengthening of lay in localized areas;

diameter reduction in localized areas; distortion of rope structure (kinking, birdcaging, crushing); steel core protrusion between the outer strands; internal corrosion; wear of outside wires; more detailed inspection of end connections for broken wires and corrosion; severely corroded, cracked, bent, worn or improperly applied end connections; waviness (corkscrew effect) of rope; high or low strand.

To establish data as a basis for judging the proper time for replacement, a dated report of rope condition at each periodic inspection must be kept on file. This report shall cover points of deterioration listed above. If the rope is replaced, only the fact that the rope was replaced need be recorded.

Certain types of ropes and applications require special attention and require reduced time intervals between periodic inspections:

- Rotation Resistant ropes have a unique construction and are susceptible to damage and increased deterioration when working under difficult conditions such as duty cycle operation.
- Boom hoist ropes because of the importance of their function and because their location may make inspection difficult.

ROPE REPLACEMENT

There are no precise rules to determine the exact time for the replacement of the rope since many variable factors are involved. Once a rope reaches any one of the removal criteria, it must be replaced immediately unless allowed to operate to the end of the work shift by the judgment of a qualified person. If the rope was not removed immediately, it shall be replaced before the end of the next work shift. Specific inspection attributes and removal criteria are:

(1) Broken wires: (a) For ropes operating on equipment covered by B30.5: In running ropes, 6 randomly distributed wire breaks per rope lay or 3 wire breaks per strand per rope lay. A rope lay is the distance that it takes one outer strand to make one complete revolution around the rope. A 6-strand rope will typically have a rope lay of 6.4 times the rope diameter (i.e. a 1/2" 6x25FW EIP IWRC RRL rope will have rope lay of 3.2") (b) For ropes operating on equipment covered by B30.2, in running ropes is 12 randomly distributed wire breaks per rope lay or four wire breaks per strand (c) For all categories of Rotation Resistant ropes, the retirement criteria is 2 wire breaks in 6 rope diameters or 4 wire breaks on 30 rope diameters (i.e. 6 rope diameters in a 1" rope is 6") (d) One broken outer wire at the contact point with the core which has worked its way out of the rope structure and protrudes, loops out or is slightly raised from the body of the rope
Note: Broken wire removal criteria cited in this volume apply to wire rope operating on steel sheaves and drums and wire rope operating on multilayer drums regardless of sheave material. Due to the difficulty in detecting wire breaks when polymer are utilized with single layer drums, the user should contact the sheave manufacturer for broken wire removal criteria.

Reductions from nominal diameter greater than 5% (Minimum Value = Nominal Diameter x .95)
Distortion of rope structure: (a) Damage resulting in distortion of the rope structure (e.g., kinking, birdcaging, crushing) (b) Steel core protrusion between the outer strands (c) Localized change in lay length (d) Changes in original geometry due to crushing forces where the diameter across the distorted section is 5/6 of the nominal diameter.

(4) Waviness (corkscrew effect) in the rope that causes overall diameter to increase to a value greater than 110% of nominal rope diameter.

(5) A high or low strand that is higher or lower than ½ of the strand diameter above or below the surface of the rope.

(6) Any apparent damage from a heat source including, but not limited to welding, power line strikes, or lightning.

(7) Widespread or localized external corrosion as evidenced by pitting, and obvious signs of internal corrosion such as magnetic debris coming from valleys.

(8) Severely corroded, cracked, bent, worn, grossly damaged, or improperly installed end connections
Note: Consult the latest edition of the ASME B30 Volume that applies to your crane as removal criteria may be updated over time based on the latest knowledge and information. All rope that has been idle for a month or more due to shut down or storage of a crane should be given a detailed inspection according to the requirements of the periodic inspection provided by the B30 standards.

ROPE SERVICE LIFE

A long-range inspection program should be established and should include records on the examination of ropes removed from service so that a relationship can be established between visual observation and actual condition of the internal structure. There are a wide variety of wire rope constructions available to be used on cranes.

It is important that the correct rope be used for each specific application. Because wire rope wears in service, the method by which the rope wears is an important factor in determining the most suitable rope. Replacement rope must have a rated strength at least equal to the original rope supplied or recommended for the machine. Any change from the original specification for the rope must be specified by the wire rope manufacturer, crane manufacturer, or qualified person. When there is a question, consult with Bridon American about the rope construction most appropriate for the application.

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